

NanoGT Match Report: Severe combined immunodeficiency due to adenosine deaminase deficiency

Disease: Severe combined immunodeficiency due to adenosine deaminase deficiency (ORPHA:277)

Primary gene: ADA

Gene CDS: 1092 bp

Inheritance: Autosomal recessive

Target tissues scored: hematopoietic

Interpretation

- At least one high-confidence precedent was found, but this is still a precedent match rather than a clinical-trial recommendation.
- Main review flags: Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility; Vector does not naturally cover all annotated disease tissues: hematopoietic; No direct tissue overlap; treat this as weak precedent unless route or modality is changed.

Disease Mechanism Evidence

Molecular mechanism: loss of function

Mechanistic detail: ADA enzyme deficiency causing toxic purine metabolite accumulation

Gene-addition compatibility: compatible

Preferred modality class: ex vivo hsc gene addition

Evidence level/status: direct / source linked needs review

Evidence summary: ADA-SCID is a deficiency disorder with an approved ex vivo autologous CD34+ ADA gene-addition precedent (Strimvelis; gammaretroviral vector)

Evidence source: [OMIM ADA-SCID 102700](#)

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
 - Modality coverage is limited mainly to AAV and integrating ex vivo HSC vector precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
 - Few catalog vectors cover the annotated disease tissue(s): LV.
 - Endpoint readiness unclear from available annotations; identify biomarkers, natural-history measures, and patient-relevant outcomes before trial prioritisation.
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Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Strimvelis	LV	8.1/10	High	approved
2	Libmeldy	LV	7.0/10	Medium	approved
3	Skysona	LV	6.8/10	Medium	approved
4	AVR-RD-01	LV	6.6/10	Medium	phase1/2
5	BMN 307	AAV5	6.4/10	Medium	phase2

Match #1: Strimvelis

Precedent disease: ADA-SCID

Vector: LV

Tissue target: hematopoietic

Composite score: 8.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.30	1.0	Immunological protection of target tissue
Promoter availability	0.70	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	8.14	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1092bp / cargo 8000bp (14% utilized)
- Vector tropism plus precedent target match: hematopoietic
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: immune_hematopoietic
- Disease mechanism: loss of function — ADA enzyme deficiency causing toxic purine metabolite accumulation
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: ADA-SCID is a deficiency disorder with an approved ex vivo autologous CD34+ ADA gene-addition precedent (Strimvelis; gammaretroviral vector)
- Mechanism source: OMIM ADA-SCID 102700 (<https://omim.org/entry/102700>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden

- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: low privilege — immune cells reside here; robust conditioning and monitoring required
- Promoter availability: EFS, PGK, SFFV — validated across integrating ex vivo HSC programs; ADA-SCID includes the gammaretroviral Strimvelis precedent, while beta-thalassaemia uses lentiviral HSC platforms
- Route of administration: Ex vivo HSC delivery — technically complex but highly precise; gold standard for blood disorders
- ORGANELLE TARGETING: COMPATIBLE — ADA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility

Match #2: Libmeldy

Precedent disease: Metachromatic leukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 7.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.30	1.0	Immunological protection of target tissue
Promoter availability	0.70	1.0	Validated tissue-specific promoters exist

Dimension	Score	Max	What it measures
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.95	10.0	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1092bp / cargo 8000bp (14% utilized)
- Vector tropism plus precedent target match: hematopoietic
- Protein class mismatch
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (immune hematopoietic vs leukodystrophy)
- Disease mechanism: loss of function — ADA enzyme deficiency causing toxic purine metabolite accumulation
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: ADA-SCID is a deficiency disorder with an approved ex vivo autologous CD34+ ADA gene-addition precedent (Strimvelis; gammaretroviral vector)
- Mechanism source: OMIM ADA-SCID 102700 (<https://omim.org/entry/102700>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: low privilege — immune cells reside here; robust conditioning and monitoring required
- Promoter availability: EFS, PGK, SFFV — validated across integrating ex vivo HSC programs; ADA-SCID includes the gammaretroviral Strimvelis precedent, while beta-thalassaemia uses lentiviral HSC platforms
- Route of administration: Ex vivo HSC delivery — technically complex but highly precise; gold standard for blood disorders
- ORGANELLE TARGETING: COMPATIBLE — ADA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility

Match #3: Skysona

Precedent disease: Cerebral adrenoleukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 6.8 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.30	1.0	Immunological protection of target tissue
Promoter availability	0.70	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.81	10.0	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1092bp / cargo 8000bp (14% utilized)
- Vector tropism plus precedent target match: hematopoietic
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Different pathway (immune hematopoietic vs peroxisomal)
- Disease mechanism: loss of function — ADA enzyme deficiency causing toxic purine metabolite accumulation
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: ADA-SCID is a deficiency disorder with an approved ex vivo autologous CD34+ ADA gene-addition precedent (Strimvelis; gammaretroviral vector)
- Mechanism source: OMIM ADA-SCID 102700 (<https://omim.org/entry/102700>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: low privilege — immune cells reside here; robust conditioning and monitoring required
- Promoter availability: EFS, PGK, SFFV — validated across integrating ex vivo HSC programs; ADA-SCID includes the gammaretroviral Strimvelis precedent, while beta-thalassaemia uses

- lentiviral HSC platforms
- Route of administration: Ex vivo HSC delivery — technically complex but highly precise; gold standard for blood disorders
- ORGANELLE TARGETING: COMPATIBLE — ADA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility

Match #4: AVR-RD-01

Precedent disease: Fabry disease

Vector: LV

Tissue target: hematopoietic

Composite score: 6.6 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.30	1.0	Immunological protection of target tissue
Promoter availability	0.70	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.57	10.0	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1092bp / cargo 8000bp (14% utilized)

- Vector tropism plus precedent target match: hematopoietic
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Different pathway (immune_hematopoietic vs lysosomal_storage)
- Disease mechanism: loss of function — ADA enzyme deficiency causing toxic purine metabolite accumulation
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: ADA-SCID is a deficiency disorder with an approved ex vivo autologous CD34+ ADA gene-addition precedent (Strimvelis; gammaretroviral vector)
- Mechanism source: OMIM ADA-SCID 102700 (<https://omim.org/entry/102700>)
- Approval status: phase1/2
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: low privilege — immune cells reside here; robust conditioning and monitoring required
- Promoter availability: EFS, PGK, SFFV — validated across integrating ex vivo HSC programs; ADA-SCID includes the gammaretroviral Strimvelis precedent, while beta-thalassaemia uses lentiviral HSC platforms
- Route of administration: Ex vivo HSC delivery — technically complex but highly precise; gold standard for blood disorders
- ORGANELLE TARGETING: COMPATIBLE — ADA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility

Match #5: BMN 307

Precedent disease: Phenylketonuria

Vector: AAV5

Tissue target: liver

Composite score: 6.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match

Dimension	Score	Max	What it measures
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.30	1.0	Immunological protection of target tissue
Promoter availability	0.70	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.43	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1092bp / cargo 4700bp (23% utilized)
- No tissue overlap (disease: ['hematopoietic'], vector: ['liver', 'lung', 'CNS'])
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (immune_hematopoietic vs amino_acid_metabolism)
- Disease mechanism: loss of function — ADA enzyme deficiency causing toxic purine metabolite accumulation
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: ADA-SCID is a deficiency disorder with an approved ex vivo autologous CD34+ ADA gene-addition precedent (Strimvelis; gammaretroviral vector)
- Mechanism source: OMIM ADA-SCID 102700 (<https://omim.org/entry/102700>)
- Approval status: phase2
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: low privilege — immune cells reside here; robust conditioning and monitoring required
- Promoter availability: EFS, PGK, SFFV — validated across integrating ex vivo HSC programs; ADA-SCID includes the gammaretroviral Strimvelis precedent, while beta-thalassaemia uses lentiviral HSC platforms
- Route of administration: Ex vivo HSC delivery — technically complex but highly precise; gold standard for blood disorders
- ORGANELLE TARGETING: COMPATIBLE — ADA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: hematopoietic
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone